

Analyzing Thematic Alignment in Scientific Journals

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1 Introduction

Every scientific journal publishes an Aims & Scope: a short document that draws a boundary around the topics the venue considers its own. Authors use it to decide where to submit, editors use it to assign reviewers, and indexing services use it to classify the venue. What happens to that boundary over time, however, is rarely monitored. Research communities evolve, new subfields emerge, and the papers that accumulate in a venue can gradually pull it away from its original definition — without anyone formally changing the document.

Catching this kind of drift requires comparing what a journal says it covers against what it actually publishes, at scale and over many years. Word-level methods fall short here: the same concept appears under dozens of different phrasings across a decade of literature, and a bag-of-words model has no way to know they are related. Transformer-based sentence embeddings trained for semantic similarity solve this — they map text into a vector space where proximity reflects meaning, not just shared vocabulary [1]. Complementary to alignment scoring, topic modeling with BERTopic [2] provides an explicit view of the conceptual composition of a corpus, enabling identification of which thematic clusters are driving any observed drift.

Prior work on journal-level thematic analysis has largely relied on citation networks and keyword co-occurrence to study disciplinary boundaries [3, 4]. This project takes a different path: instead of metadata, it operates directly on the semantic content of abstracts, enabling fine-grained, sentence-level comparison between each paper and the journal's declared mission. The rest of this report is organized as follows. Section 2 states the research question and methodology. Section 3 presents experimental results. Section 4 provides a critical discussion and directions for future work.

2 Research Question and Methodology

The central research question of this project is: Can we quantitatively detect thematic drift in a scientific journal by measuring the semantic alignment between its published papers and its stated Aims & Scope — and if so, which topics are driving the change?

We formalize the problem as follows. Let $S = \{s_1, s_2, \dots, s_m\}$ be the set of sentences in the journal's Aims & Scope document, and let $P = \{p_1, p_2, \dots, p_n\}$ be the set of paper abstracts in the corpus. For each paper p_i , we define its alignment score $A(p_i)$ as the cosine similarity between its abstract embedding and a single Aims & Scope reference vector. Thematic drift is then operationalized as a downward trend in the annual mean of $A(p_i)$ over time. Outliers are defined as papers in the bottom 5th percentile of the score distribution.

2.1 Data Collection

Papers were collected from the arXiv API using the cs.CL (Computational Linguistics) category filter, covering 2015–2026, yielding a final corpus of 1,918 abstracts with balanced yearly sampling (150–193 papers per year). The official Aims & Scope of the Computational Linguistics journal was downloaded directly from the publisher's website and saved as a fixed reference snapshot to ensure reproducibility.

2.2 Content Modeling and Alignment Scoring

Both the Aims & Scope text and all abstracts were encoded using Sentence-BERT (all-MiniLM-L6-v2) [1], producing 384-dimensional L2-normalized vectors. Cosine similarity was computed between each abstract embedding and every scope sentence embedding. The final alignment score per paper is the cosine similarity between its abstract embedding and the single Aims & Scope reference vector. This straightforward formulation is efficient and consistent across all papers in the corpus.

2.3 Topic Modeling

BERTopic [2] was applied to the full abstract corpus to discover latent thematic clusters without requiring a pre-specified topic count. The model combines UMAP dimensionality reduction, HDBSCAN clustering, and class-based TF-IDF topic descriptors to produce interpretable topic labels. Each identified topic was then assigned a mean alignment score to assess

which thematic areas contribute most to misalignment. A model comparison experiment additionally evaluated three encoders — all-MiniLM-L6-v2, all-mpnet-base-v2, and allenai-specter — to verify robustness of alignment scores across encoder choices.

3 Experimental Results

3.1 Dataset

The final corpus contains 1,918 abstracts published in the arXiv cs.CL category between 2015 and 2026. Yearly counts range from 150 to 193 papers. The global alignment score distribution has mean = 0.282, std = 0.084, min = −0.004, max = 0.566, and median = 0.284.

3.2 Alignment Score Distribution and Drift

The global distribution is unimodal, centered between 0.20 and 0.36 (Figure 1). There is no bimodal structure, indicating a continuous spectrum of alignment rather than a clean split between on- and off-scope papers. Alignment peaked in 2016 (mean = 0.306) and declined steadily to 0.237 by 2026 — a relative drop of ~22% (Figure 2). The percentile trend (p10/median/p90) confirms that this decline is consistent across the full distribution rather than being driven exclusively by new extreme outliers.

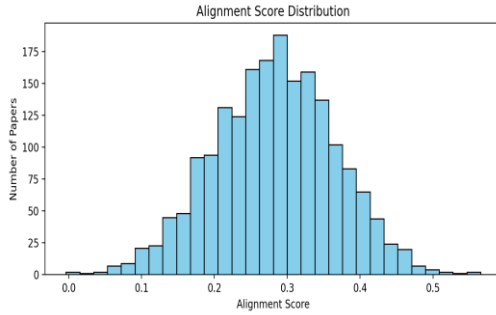


Figure 1: Global alignment score distribution.

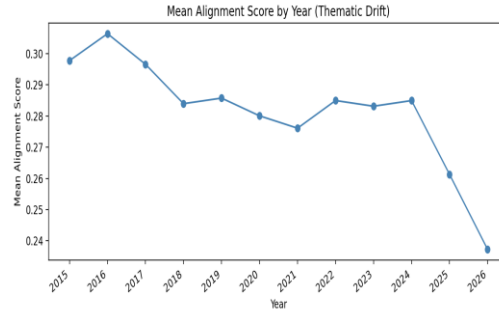


Figure 2: Yearly mean alignment score (2015–2026).

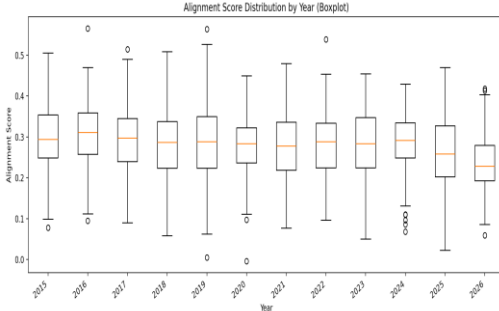


Figure 3: Score distribution by year.

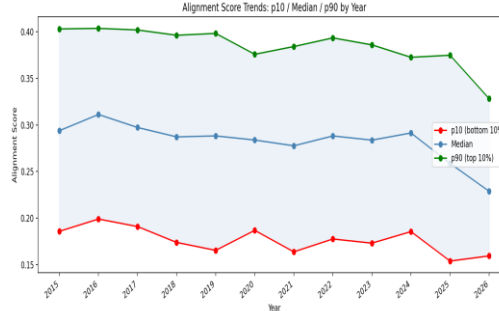


Figure 4: Percentile trends (p10 / median / p90).

Table 1 provides year-by-year summary statistics. Notable observations include the relatively stable period 2018–2024 (mean scores between 0.276 and 0.286) and the sharp drop in 2025–2026, which coincides with a surge of LLM-related submissions spanning computer vision, audio, and systems research.

Table 1: Year-wise alignment statistics.

Year	Mean Score	Std Dev	Outliers	Rate
2015	0.298	0.085	7	4.5%
2016	0.306	0.079	4	2.4%
2017	0.297	0.081	5	2.9%
2018	0.284	0.087	9	4.9%
2019	0.286	0.092	10	5.2%

2020	0.280	0.077	7	4.7%
2021	0.276	0.086	11	7.3%
2022	0.285	0.081	6	4.0%
2023	0.283	0.084	8	5.3%
2024	0.285	0.071	7	4.7%
2025	0.261	0.087	12	8.0%
2026	0.237	0.068	10	6.7%

3.3 Outlier Analysis

96 papers (5.0%) fall below the outlier threshold of 0.1435. Qualitative inspection of the lowest-scoring papers confirms the metric is meaningful: the three most misaligned papers are "Measuring University Impact: Wikipedia approach" (score: -0.004), "Jointly optimal dereverberation and beamforming" (0.005), and "FLowHigh: Towards Efficient and High-Quality Audio Super-Resolution" (0.023) — none of which belong to computational linguistics. The year-wise outlier rate (Figure 5) shows pronounced spikes in 2021 (7.3%) and 2025 (8.0%), consistent with increasing category heterogeneity in recent years.

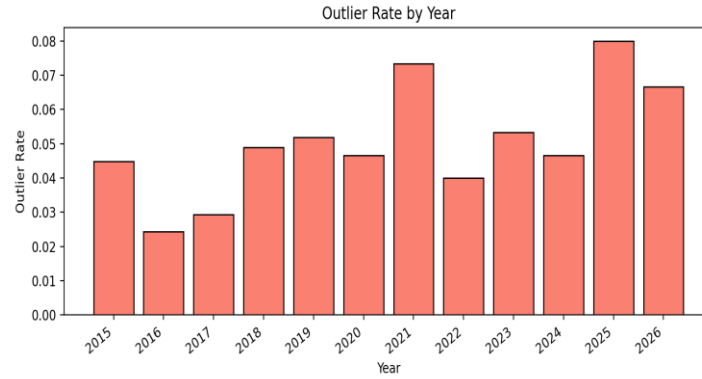


Figure 5: Outlier rate by year (bottom 5% threshold).

3.4 Topic Structure and Conceptual Evolution

BERTopic identified coherent topic clusters with clearly differentiated mean alignment scores. The most aligned topics correspond to classical NLP tasks: dependency parsing, natural language generation, and morphological analysis. The least aligned topics — image/video processing, audio super-resolution, adversarial machine learning — reflect content that has migrated into the cs.CL category but falls outside the journal's scope. 38% of papers were assigned to Topic -1 (no coherent cluster), reflecting the breadth of the category. The UMAP projection (Figure 7) visually confirms this separation: papers close to the Aims & Scope reference (★) are consistently higher-scoring.

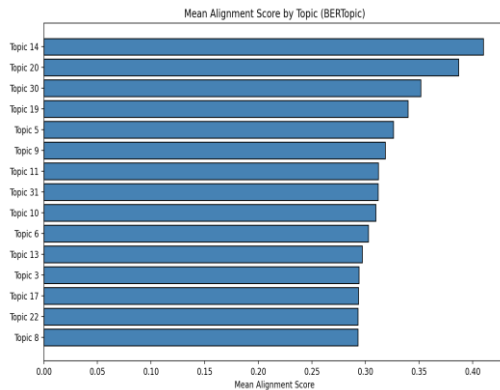


Figure 6: Mean alignment score by BERTopic topic.

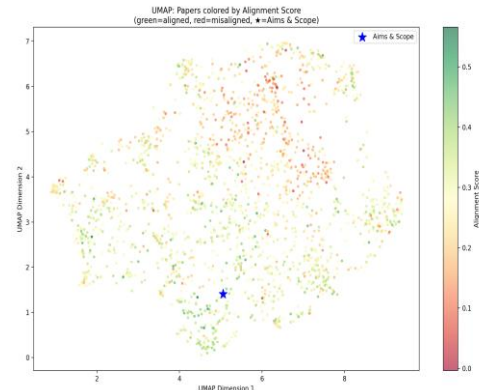


Figure 7: UMAP colored by alignment score (★ = Aims & Scope).

4 Concluding Remarks

This project demonstrated that thematic drift in a scientific journal can be quantitatively detected and characterized using sentence embeddings and cosine similarity alignment scores. The results for the arXiv cs.CL category (2015–2026) are clear: alignment peaked in 2016 and has declined by ~22%, with outlier rates rising sharply in 2021 and 2025. Topic modeling confirmed that the decline is driven by identifiable thematic clusters — primarily audio, image processing, and systems work — that have grown in the category but fall outside the Computational Linguistics journal scope.

Critically, a mean score of 0.282 indicates only moderate alignment even at the corpus level. This raises an important interpretive question: is the Aims & Scope itself sufficiently specific, or does a broad scope statement naturally produce moderate scores regardless of drift? A controlled comparison against a venue with a narrower scope would help establish a meaningful baseline. Additionally, this pipeline operates only on abstracts, which may not fully represent a paper's content; full-text analysis could reveal different alignment patterns for theory-heavy papers whose contributions are not well-summarized by their abstracts.

The encoder choice was validated through a three-model comparison (MiniLM, MPNet, SPECTER), which showed consistent distributional trends across all three, supporting the robustness of the findings. However, SPECTER's domain-specific fine-tuning produced somewhat different absolute score ranges, suggesting that score thresholds (such as the outlier cutoff) should be recalibrated when switching encoders.

Future extensions of this work could include: (i) full-text analysis beyond abstracts; (ii) citation-based journal identity validation to reduce arXiv category noise; (iii) fine-grained sentence-level alignment to identify which specific scope dimensions are eroding; and (iv) application to multiple journals simultaneously to enable cross-venue comparison of thematic stability.

AI Usage Disclaimer

Parts of this project were developed with the assistance of Anthropic's Claude (claude-sonnet-4-6). The AI was used to support the structuring of the report, drafting and refining descriptive text, and formatting the final document. All content produced with AI assistance has been carefully reviewed, edited, and validated by me. The research question, methodology design, code implementation, and interpretation of results are entirely my own work. I take full responsibility for the final content and its accuracy, relevance, and academic integrity.

References

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